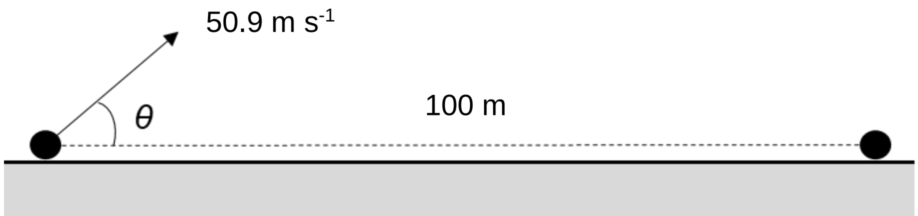
2	An object is launched with a speed of 50.9 m s^{-1} at an angle θ above the horizontal, as shown.  The ground is level. The object lands on the ground 100 m from its initial launch position. What is the value of the angle θ? You may need to make use of the double angle formula: $\sin(2\theta) = 2\sin\theta\cos\theta$						
	A	2.8°	**B**	5.5°	**C**	11°	**D**	79°
	Solution: C In the horizontal direction: $50.9\, t \cos\theta = 100$ In the vertical direction (taking upwards as positive): $0 = 50.9\, t \sin\theta - (0.5)(9.81)t^2 = 50.9 \sin\theta - \tfrac{1}{2}(9.81)t$ $50.9 \sin\theta = (0.5)(9.81)t$ $t = \dfrac{50.9\sin\theta}{(0.5)(9.81)}$						

Substituting into the previous equation:

$$100 = \frac{50.9^2 \sin\theta \cos\theta}{(0.5)(9.81)} = \frac{50.9^2 (0.5)(\sin 2\theta)}{(0.5)(9.81)}$$

$$2\theta = 22^\circ$$

$$\theta = 11^\circ$$

3	Two spheres with different masses are initially moving towards each other with the same